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Synthesizing True Golden Voices to Enhance Pronunciation Training for Individual Language Learners

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Abstract

A question that has received researchers' attention in pronunciation training is: Who might be an ideal, or "golden" speaker for a learner to imitate? Previous studies have explored the voices that blend model pronunciation with the learner's voice quality. However, a well-known phenomenon called voice confrontation occurs when one's recorded voice sounds different from how s/he perceives it. This discrepancy arises because self-perceived voices (SPVs) contain both air-conducted and body-conducted components. In this study, native speech samples are converted to match learners' recorded voice quality, which is then further adjusted to resemble their SPVs using voice conversion techniques. Speech imitation training and assessment experiments reveal that learners prefer their SPV quality and imitate target pronunciation more accurately with it than with other voice qualities tested, such as their recorded voices and the voices of other speakers.

Index Terms: Language learning, speech imitation, golden voice, voice confrontation, self-perceived voice

1. Introduction

In studies of foreign language learning, researchers often claim that learners do not need to acquire a native-sounding pronunciation as long as their speech is sufficiently intelligible [1–4]. However, certain groups of learners still strive to achieve native pronunciation. For example, many non-native teachers aim to acquire a model pronunciation close to native varieties such as General American (GA) or Received Pronunciation (RP). Non-native members of English drama clubs seek native-sounding pronunciation for their performances, and even native actors and actresses may need to master regional accents precisely to succeed in auditions. In these cases, learners aim to accurately imitate the target pronunciation or accent [5].

Pronunciation training can be seen as mastering the articulation of vocal organs to produce speech sounds with the required phonetic quality, typically achieved by listening to model speech samples. However, the timbre and pitch of a teacher's or trainer's voice generally differ from those of a learner or trainee. In vocal training for singing, a melody is often played on a piano, with its key or baseline pitch adjusted to match the trainee's baseline pitch [6]. In contrast, in speech training for language learning, adjusting the timbre and pitch of a teacher's voice is difficult or impossible, as the teacher is not a machine. With advances in speech technologies, machine teachers have been developed as part of CALL (Computer-Aided Language Learning) systems [7], raising an important question: Who is the ideal, or golden, speaker for a specific learner to imitate? [8, 9]

In brain sciences, it is well established that listening to speech can involuntarily activate motor regions responsible for articulation—a phenomenon known as mirroring [10–12]. This mechanism suggests that learners' brains respond to heard speech by internally simulating the associated articulatory gestures. From this perspective, the question of optimal model pronunciation can be reframed as: Whose voice most effectively stimulates a learner's mirror neurons? Since articulatory control is most naturally aligned with the speaker's own voice, it is reasonable to hypothesize that the most effective model would be a native speech sample acoustically modified to match the learner's internal voice characteristics—specifically, their SPV.

Previous studies have explored native speech adapted to more closely resemble the recorded voice timbre or pitch of individual learners [8, 9, 13]. However, such studies relied on the learner's recorded voice—i.e., air-conducted voice (ACV)—as a reference. As highlighted in studies on voice confrontation [14–16], individuals often experience discomfort or disassociation when hearing recordings of their own voices, largely due to a mismatch between the ACV and their internally heard SPV. The SPV contains both air-conducted and body-conducted components [17–20], and cannot be directly measured, as it is shaped by physical and perceptual factors and experienced subjectively. Because of this, adapting native speech samples to match a learner's SPV remains a fundamental challenge in personalized pronunciation training[1].

Recent research has proposed methods for approximating the body-conducted component using multiple sensing devices [21, 22]. By combining the estimated body-conducted signals with air-conducted recordings at appropriate weights, it becomes possible to emulate a learner's SPV more realistically. In this study, we integrate such emulation with state-of-the-art speech-to-speech voice conversion techniques to transform native speech samples into what we term the true golden voice (TGV)—native speech that has been acoustically shaped to closely resemble the timbre and pitch of the learner's SPV. Our goal is to examine whether the TGV enables learners to imitate native speech more effectively and naturally, and to explore its potential for personalized language learning applications.

The scientific contributions of this paper are as follows:

- To the best of our knowledge, this study is the first to synthesize native speech samples matched to the voice timbre and pitch of learners' SPVs—referred to as TGVs—for pronunciation training.
- Both subjective and objective assessments of the true golden voices are conducted experimentally with Japanese learners of English, some of whom are language teachers, as well as amateur or semi-professional actors and actresses.

[1]Given the transmission of vibrations through bones and internal tissues from the vocal organs to the inner ear, the fundamental frequencies of recorded and self-perceived voices are essentially identical.

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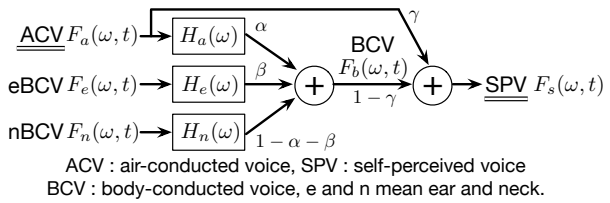

Figure 1: *BCV as weighted summation of ACV, eBCV, and nBCV*

2. Techniques Applied for the Golden Voices

2.1. Speech-to-speech and text-to-speech model

Recently, many text processing models have become accessible for educational and research use, either freely or at low cost, including Large Language Models (LLMs) [23–26]. Among speech generation technologies, ElevenLabs provides a model capable of both speech-to-speech (STS) and text-to-speech (TTS) conversion [27]. This model requires approximately 30 minutes of speech samples in the target language to build a personalized voice model, without the need for transcripts. Once trained, the model can generate speech outputs that replicate the speaker's voice timbre and pitch with high naturalness and quality, while retaining the pronunciation and temporal characteristics of any given input speech, such as speaking rate.

The STS/TTS model can be used to transform native speech samples by modifying them to match the target speaker's vocal characteristics, while preserving the linguistic content and prosodic features. Additionally, when a text transcript is input, the model can synthesize speech that resembles the target speaker reading the text with a consistent voice quality. The output's timing and speech rhythm are determined automatically based on the speaker's training data. Compared to other existing tools, the voice quality and functionality of ElevenLabs models have been recognized as particularly high by educators and practitioners in the field of language learning [28].

2.2. Conversion from recorded voices to self-perceived voices using multiple sensing devices

It is intuitive that SPVs can be emulated by summing ACVs and body-conducted voices (BCVs) with appropriate weighting. However, estimating BCV is challenging because they are a mental phenomenon and cannot be measured acoustically. Moreover, recent studies [29, 30] have shown that multiple vibration transmission pathways exist from the vocal organs to the inner ear, complicating BCV estimation. A method addressing this challenge using multiple sensing devices has been proposed in [21, 22], where BCVs are estimated via a weighted summation of vibrations transmitted through different pathways.

As illustrated in Figure 1, three sensing devices are employed: one for ACV, and two for detecting bone and skin vibrations—around the ear (eBCV) and neck (nBCV). These signals are weighted and combined to emulate the BCV. The emulated BCV was first validated through subjective listening tests using earmuffs[2] (Figure 2) that blocked ACV, allowing participants to compare it with their perceived BCV. Subsequently, the emulated SPV was obtained by summing the validated BCV and ACV, and was evaluated without earmuffs through comparison with the SPV participants normally hear during speech.

Six parameters shown in Figure 1—three transfer functions $H_x(\omega)$ for $x = a, e, n$, and three weights α, β, and γ—were optimized through listening-based fine-tuning. To reduce time cost, an Interactive Genetic Algorithm was employed in [22]. However, this method produces only a synchronized parallel corpus between (ACV, eBCV, nBCV) and the emulated BCV or SPV. For practical pronunciation training, direct conversion from ACV to emulated SPV is necessary, as underlined in Figure 1. This capability was implemented in the current study using GMM-based conversion, described in the next section.

[2]The earmuffs, lined with sound-absorbing urethane sponge on all inner surfaces, effectively blocked ACV and suppressed extra standing waves, enabling isolated evaluation of the emulated BCV.

Figure 2: *Earmuffs with wired earphones installed*

3. Synthesis of the True Golden Voices

3.1. Participants

Participants were recruited through the sixth author, who provides online speech training and lectures on AI-based English education. Fifteen native Japanese speakers participated: nine females (two teenagers, two in their twenties, three in their fifties, and two in their sixties) and six males (five in their twenties and one in his forties). The group included four English teachers, four members of an English drama club, and seven adult learners from various professional backgrounds. The English teachers and drama club members were generally more accustomed to hearing their recorded voices.

3.2. STS Model Construction

To construct an STS model for each participant, they were asked to read aloud 60 short texts, each about 30 seconds in duration, as illustrated in Figure 3. Using these models, several speech samples were synthesized and presented to the participants to evaluate whether the generated voices were perceived as sufficiently natural and representative of their own recorded speech.

As noted in Section 1, the STS/TTS model generates output reflecting the voice timbre and pitch of the speaker's recorded voice. Participants often expressed surprise at the high naturalness of the generated voices, as also mentioned by language teachers in Section 2.1. However, they also noted perceptible differences between the generated samples and their SPVs.

3.3. Conversion to self-perceived voices

As described in Section 2.2, a synchronized parallel corpus between the ACV and the emulated SPV was created for each participant as follows. Three sensing devices were used: Apple wired EarPods for ACV, a SONY ECM-TL3 [31] for eBCV, and a NANZU SH-12jK [32] for nBCV, as shown in Figure 1. Participants wore these devices and read aloud simple English passages for 10 minutes in a soundproof room. A 30-second segment was extracted from each recording to fine-tune six parameters per participant[3], which were then applied to generate the full 10-minute synchronized corpus between ACV and emulated SPV, illustrated in Figure 3.

[3]Parameter tuning was repeated until participants judged the emulated SPV to closely resemble their perceived internal voice. This subjective adjustment process typically required around 10 minutes.

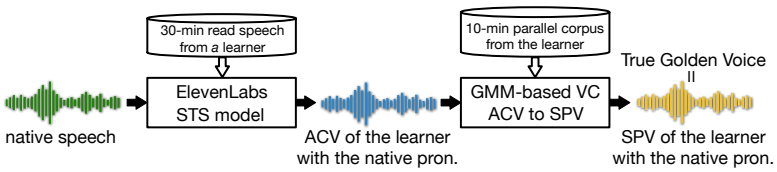

Figure 3: *Overall procedure of synthesizing the True Golden Voice (TGV) from a native speech sample*

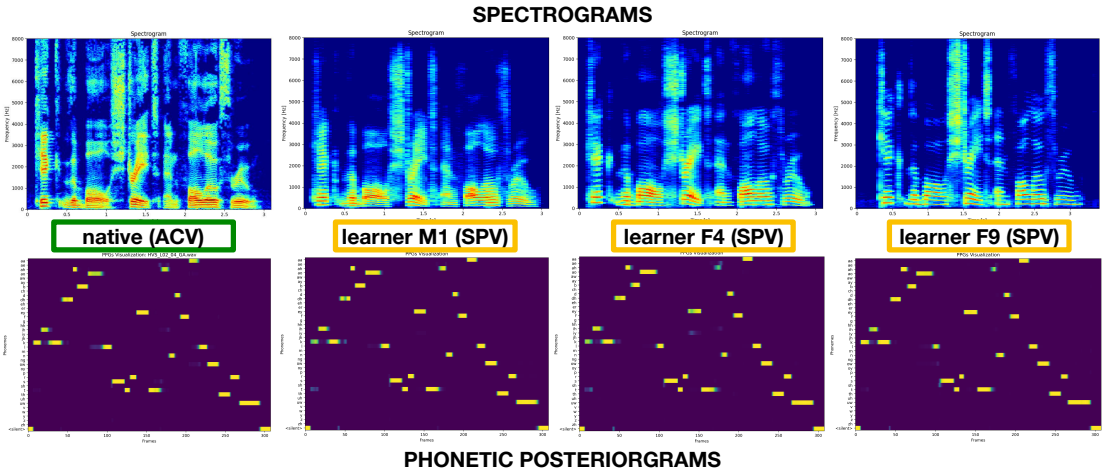

Figure 4: *Spectrograms and phonetic posteriorgrams (PPGs) computed from a native sample and the TGVs of three learners. High similarity in PPGs indicates that the phonetic content is well preserved after voice conversion.*

The conversion from ACV to SPV generally amplifies low-frequency energy, though the degree of enhancement varies notably across speakers. Technically, this was implemented via speaker-dependent GMM-based conversion [33]. Some artifacts were perceivable in the converted speech—likely due to the vocoding process—compared to the STS output.

Figure 3 shows the full procedure for converting native speech into a learner's SPV. The learner's STS model was first used to match the timbre and pitch of the native speech to their ACV, then further converted to reflect their SPV timbre. Figure 4 presents spectrograms and phonetic posteriorgrams (PPGs) [34] for a native sample and its SPV versions for learners M1, F4, and F9. The spectrograms show distinct acoustic differences across the four samples, while the PPGs indicate near-identical phonetic content. In the next section, the SPVs are evaluated for their effectiveness in pronunciation training.

4. Assessment of the True Golden Voices

4.1. Collection of GA model speech samples

In Japan, GA English is commonly used as the model pronunciation in English classes. We recorded speech samples from an assistant language teacher (ALT), a native speaker of American English in his twenties. The ALT read aloud two sets of Harvard Sentences (HS) [35], each containing ten short, phonetically balanced sentences. These sentences are widely used for standardized telephone testing and psycholinguistic studies.

To generate GA samples with a variety of speaker identities, we applied the fifteen STS models, each trained on speech data from one of the fifteen participants, to the ALT's GA samples. This process produced the GA samples with the voice timbre and pitch characteristics of the fifteen participants' ACV. Additionally, for each participant, we converted their GA samples to match the voice quality of their SPV. As a result, we created a total of 600 GA samples – half with the timbre and pitch of the participants' ACV and the other half with those of their SPV.

4.2. Imitation training for the GA pronunciation

Imitation training using the above model samples was conducted based on the speech training provided by the sixth author. A model sample was visually presented to a participant on a web-based platform, displaying its transcript and waveform. By clicking the Start Playing button, the sample was played back while an indicator moved along the waveform, allowing the participant to see exactly when the spoken content began. The participant was instructed to repeat and imitate the model sample as precisely as possible while listening to it, aiming to temporally overlap the two utterances. After recording, three playback options were available: Play Model, Play Mine, and Play Both. In Play Both, the model speech and the participant's recording were played separately in the left and right audio channels, enabling an easy comparison of how well the two utterances aligned in terms of articulatory and prosodic control. If the model speech had the timbre and pitch of the participant's SPV, his/her recording was converted to SPV before playback.

4.3. Experimental conditions of the final recording

After practicing with the model samples on the platform as many times as they desired, the participants performed three final recordings, during which only the transcript was presented.

For each participant, twenty model GA samples were prepared, and their voice qualities are categorized into four types:

1. ACV of the participant (P-ACV)
2. SPV of the participant (P-SPV)
3. ACV of others of the same gender (SG-ACV)
4. ACV of others of the different gender (DG-ACV).

Each type contained five samples. For the third and fourth types, two sets of five different speakers were used. The assignment of the HS sentences to each type was fixed across all the participants. However, the presentation order of the 20 model samples was randomly varied among the participants. We informed them in advance that the four types of model samples were to

be presented in terms of their timbre and pitch characteristics, but the type used for each presentation was not disclosed.

4.4. Subjective and objective assessment

4.4.1. Subjective assessment

After all the recordings were completed, three questionnaires were carried out, asking the participants to rank the four types of the voice timbre and pitch based on the following criteria (ranking: 1 = easiest, 4 = hardest):

A. Ease of imitation

B. Ease of comparison.

Further, we asked the participants to judge which type they preferred for future imitation training out of the three types, P-ACV, P-SPV, and others (direct use of native voices).

4.4.2. Objective assessment

Similarity in articulatory and prosodic control was objectively quantified between the model samples and the participants' recordings. All the utterances were converted into their PPGs, and dynamic time warping (DTW) was performed for every pair of model and participant utterances [34, 36]. Forced alignment (FA) using the Kaldi ASR [37] was applied to the model samples to detect spoken segments. Using the optimal paths from PPG-DTW and the FA features, the mean PPG-DTW local distance over all the spoken frames was calculated for each participant to quantify the similarity in articulatory control, separately for each of the four timbre and pitch types.

For prosodic control, we focused on pitch control. Along the optimal PPG-DTW paths, we extracted all the frame pairs that were judged to be voiced. The correlation between the fundamental frequencies of the model samples and those of the participants' overlapped recordings was then calculated for each of the four types to quantify the similarity in prosodic control.

4.5. Results and discussion

The participants ranked the four types of timbre and pitch (P-ACV, P-SPV, SG-ACV, and DG-ACV) in terms of ease of imitation and comparison. Table 1 presents the average rankings. Additionally, the participants selected their preferred type for future training, with the number of votes shown in the table.

The ranking results clearly indicate that P-ACV and P-SPV are significantly easier to imitate and compare, while DG-ACV is significantly more difficult, at least from a subjective perspective. These trends are consistent with findings from prior studies on pronunciation modeling [8, 9]. Although the rankings for P-ACV and P-SPV are similar both in imitation and comparison, for future training, ten out of the fifteen participants selected P-SPV, while the remaining five chose P-ACV.

Additionally, we examined whether preferences varied by gender, professional background (e.g., language teachers or actors), or age group, but no clear trends in preference were observed across these attributes. Post-experiment interviews revealed that participants who preferred P-SPV often cited its strong familiarity and naturalness. In contrast, those who chose P-ACV noted minor artifacts in the SPV, likely caused by the GMM-based conversion process described in Section 3.3. As noted in Section 3.1, eight of the participants were language teachers, actors, or actresses, who are presumably accustomed to hearing their ACVs. If the experiment were conducted with a larger group of ordinary language learners, the preference for P-SPV would likely be even stronger.

Table 1: *Subjective scores for each of the four types*

task	P-ACV	P-SPV	SG-ACV	DG-ACV
imitation↓	1.67	1.40	2.93	4.00
comparison↓	1.80	1.87	2.80	3.53
	P-ACV	P-SPV	others	
preference↑	5	10	0	

Table 2: *Objective scores for each of the four types*

	P-ACV	P-SPV	SG-ACV	DG-ACV
PPG↓	0.270	0.254	0.250	0.272
pitch↑	0.756		0.697	0.709

The benefits of P-SPV were also confirmed in the objective assessment. Table 2 shows the average scores for the four types. In the PPG analysis, participants' recordings in P-SPV and SG-ACV showed significantly higher phonetic similarity to the model samples than those in P-ACV and DG-ACV ($p<0.01$, frame-based t-test). This may reflect a form of voice confrontation: while P-ACV preserves the naturalness of human speech, some participants perceived it as acoustically distant from their internal voice image, potentially reducing imitation accuracy.

In the pitch analysis, as the GMM-based conversion altered only timbre, P-ACV and P-SPV were combined into a single class. As shown in Table 2, they significantly enhanced pitch imitation compared to SG-ACV and DG-ACV ($p<0.01$, utterance-based t-test). This result aligns with typical vocal training practice, such as piano key tuning, where adjusting the model pitch to the trainee's baseline improves pitch accuracy.

Although the experiments demonstrated several strong benefits of P-SPV for pronunciation training, some limitations should be acknowledged. First, given the participants' learning backgrounds, some were already advanced learners of English who were highly motivated to further improve their pronunciation. To generalize the findings, future studies should include a broader range of learners, particularly those at beginner or intermediate proficiency levels. Second, the SPVs used in this study were generated via a two-stage conversion process, as illustrated in Figure 3. By implementing neural voice conversion to directly map native samples to the SPVs, the artifacts detected by the participants can be eliminated. Lastly, in this study, GA was adopted as the model pronunciation. If more specific accents that are unfamiliar to learners were used instead, the effects of SPV might emerge after a longer period of practice.

5. Conclusions and future work

In this paper, we applied state-of-the-art voice conversion techniques to enhance pronunciation training for individual learners. Native samples were transformed to match the voice timbre and pitch characteristics of each learner's SPV, referred to as the TGV. Compared to conventional ACV-based models, the SPV-based TGV offers a more familiar and cognitively efficient input, improving articulatory and prosodic imitation. Assessment experiments demonstrated that learners imitated the model pronunciation more accurately with the TGV.

This study highlights the theoretical and practical value of SPV in personalised voice training, where model voices align with learners' internal auditory expectations. The findings contribute to the development of adaptive CALL systems across languages, accents, and learner profiles. For future work, we will explore broader functions of the TGV. By preparing samples of World Englishes using the TGV, learners may more easily adapt to these varieties, thereby improving their ability to understand diverse global pronunciations.

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